Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/24/2006
PI Name:	Dinsmore, Anthony Ph.D.		
Project Title:	Gelatin of Colloidal Particles on Droplet Surfaces, Dimension Curvature, and Droplet Elasticity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Colloid science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01003-9300	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	09/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Levine, Alex (University of Massachusetts, Amherst)		
Grant/Contract No.:	NNC05GA40G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this proposal is to elucidate the rheological effects of small particles aggregating on the surfaces of suspended liquid droplets. In general, the elasticity, stability, and coalescence of liquid droplets are central to many microgravity operations and industrial processes. Droplets are commonly used in waste removal, fire suppression, heat transfer, encapsulation, coatings deposition, and mineral purification. Moreover, droplet dynamics play a key role in the boiling of liquids in microgravity. In these examples, the surface tension and surface viscoelasticity determine the droplet behavior. In particular, the behavior of liquid droplets containing small suspended particles opens a number of fascinating fundamental questions in complex fluids, as well as a number of technological challenges. Many fluids of interest – paint, coffee, blood, lubricating oil, and dusty air – contain suspended particles. The primary significance of these particles is that they tend to adsorb at the droplet surfaces and modify the liquid/gas interfacial energy. They may make the interface viscoelastic or even rigid and, for example, suppress droplet coalescence. We propose a series of		

	experiments, simulations, and analytical modeling to investigate the gelation of particles on curved surfaces, the viscoelasticity of the resulting surface gel (or glass), and the elasticity and stability of the droplet as a whole. The results are expected to contribute to a broader understanding of random solids generally as well as a set of problems with direct relevance to microgravity operations, and may lead to future flight experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The goal of this proposal is to measure and understand the rheological effects of small particles aggregating on the surfaces of liquid droplets in suspension. The elasticity, stability, and coalescence of liquid droplets are central to many microgravity operations and industrial processes. Droplets are commonly used in waste removal, fire suppression, heat transfer, encapsulation, coatings deposition, and mineral purification. Moreover, droplet dynamics play a key role in the boiling of liquids in microgravity. In these examples, the surface tension and surface viscoelasticity determine the droplet behavior. In particular, the behavior of liquid droplets containing small suspended particles opens a number of fascinating fundamental questions in complex fluids, as well as a number of technological challenges. Many fluids of interest – paint, coffee, blood, lubricating oil, and dusty air – contain suspended particles. Here we study the effects of adsorption of such particles at the droplet surfaces. The adsorption makes the interface viscoelastic or even rigid and, for example, suppress droplet coalescence. Experiments to date have provided measurements of the interactions among adsorbed particles at flat and at curved surfaces. This topic has been the subject of controversy in the literature and the nature of our data will constitute a valuable contribution. Using simulations and analytical modeling, we also investigate the aggregation of particles on curved surfaces and the viscoelasticity of the resulting surface gel as well as the elasticity of the droplet as a whole. In addition, a combination of experiments and calculations have provided new insights into the hydrodynamics (friction and flow fields) of particles moving at liquid interfaces. Finally, simulations have addressed the limiting case in which the particles constitute the majority phase. The results are expected to contribute to a broader understanding of random solids generally as well as a set of problems with direct relevance to microgravity operations.
Bibliography Type:	Description: (Last Updated: 12/15/2010)
Articles in Peer-reviewed Journals	Brewster R, Grest GS, Landry JW, Levine AJ. "Plug flow and the breakdown of Bagnold scaling in cohesive granular flows." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Dec;72(6 Pt 1):061301. PMID: 16485940 , Dec-2005
Articles in Peer-reviewed Journals	Zeng C, Bissig H, Dinsmore AD. "Particles on droplets: From fundamental physics to novel materials." Solid State Communications, in press, May 2006. , May-2006